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1. f is a function that satisfies the following:
   - f is in $O(n^2)$,
   - f is in $\Omega(n)$,
   - f is neither in $\theta(n)$ nor in $\theta(n^2)$.

   Can you give an example of such a function f? Show that the function you name indeed satisfies all of the above.

$$n < f(n) < n^2$$

$$f(n) = n * \log(n)$$

Condition 1	Condition 2	Condition 3	Condition 4
$f\ is\ in\ O(n^2)$	$f\ is\ in\ \omega(n)$	$f\ is\ not\ in\ \ominus(n)$	$f\ is\ not\ in\ \ominus(n^2)$
$n\log(n) \le c * n^2, \forall n \ge n_o$	$nlog(n) \ge c * n, \forall n \ge n_0$	$\lim_{n\to\infty} \frac{nlog(n)}{n} = \lim_{n\to\infty} \log(n) = \infty$	$\lim_{n\to\infty} \frac{nlogn}{n^2} = \lim_{n\to\infty} \frac{logn}{n}$
$Since\ \log(n) < n, \forall n > 0$	$\log(n) \ge c$		Using L's Hôpital Rule: $\lim_{n\to\infty} \frac{\frac{1}{n}}{1} = 0$
$nlog(n) < n * n$	For any $n \ge 2$ (Pick 2 due to base 2 for the log), $logn \ge 1$	Because the limit is infinity, f(n) grows strictly faster than n	Because the limit is zero, f(n) grows strictly slower than n^2
This hold for $c = 1, n_0 = 1$. Thus $f(n) \in \Omega(n^2)$	This holds for $c = 1, n_0 = 2$. Thus, $f(n) \epsilon\ \Omega(n)$	Therefore, $f(n) \notin \ominus(n)$	Therefore, $f(n) \notin \ominus(n^2)$

2. For each pair of functions given below, point out the asymptotic relationships that apply: $f = O(g)$, $f = \theta(g)$, and $f = \Omega(g)$.

- $f(n) = \sqrt{n}$ and $g(n) = n$
- $f(n) = \log n$ and $g(n) = \sqrt{n}$
- $f(n) = 1$ and $g(n) = 2$
- $f(n) = 100 \cdot 2^n$ and $g(n) = 3^n$
- $f(n) = 2^{2n+2}$ and $g(n) = 4^{n+3}$
- $f(n) = 10n \cdot log\, n$ and $g(n) = n \cdot log\, 10n$
- $f(n) = n!$ and $g(n) = (n+1)!$

$$1.\, f(n) = \sqrt{n}\ and\ g(n) = n$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} = \frac{1}{0} = 0$$

$$f = O(g)$$

$$2.\, f(n) = \log(n)\ and\ g(n) = \sqrt{n}$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{\log(n)}{\sqrt{n}}.\, Using\ L's\ Hopital\ Rule$$

$$\lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} = \lim_{n\to\infty} \frac{2\sqrt{n}}{n} = \lim_{n\to\infty} \frac{2}{\sqrt{n}} = 0$$

$$f = O(g)$$

$$3.\, f(1) = 1\ and\ g(n) = 2$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{1}{2} = 0.5$$

$$f = \ominus\ (g)$$

$$4.\, f(n) = 100 * 2^n\ and\ g(n) = 3^n$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{100 * 2^n}{3^n} = 100 * \left(\frac{2}{3}\right)^n .\, Since\, \frac{2}{3} < 1$$

$$100 * 0 = 0$$

$$f = O(g)$$

$$5.\, f(n) = 2^{2n+2}\ and\ g(n) = 4^{(n+3)}$$

$$f(n) = 2^2 * (2^2)^n = 4 * 4^n$$

$$g(n) = 4^3 * 4^n = 64 * 4^n$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{4 * 4^n}{64 * 4^n} = \frac{4}{64} = \frac{1}{16}$$

$$f = \ominus\ (g)$$

$g(n): \log(10n) = \log(10) + logn. So, g(n) = nlogn + nlog(10)$ $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{10nlogn}{nlogn + nlog10} = nlogn * \frac{10}{1 + \frac{log10}{logn}} = 10 + (1 + 0) = 10$ $f = \ominus (g)$
$7. f(n) = n!\, and\ g(n) = (n+1)!$ $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{n!}{(n+1)!} = \frac{n!}{(n+1) * n!} = \frac{1}{n+1}$ $\lim_{n\to\infty} \frac{1}{n+1} = 0$ $f = O(g)$